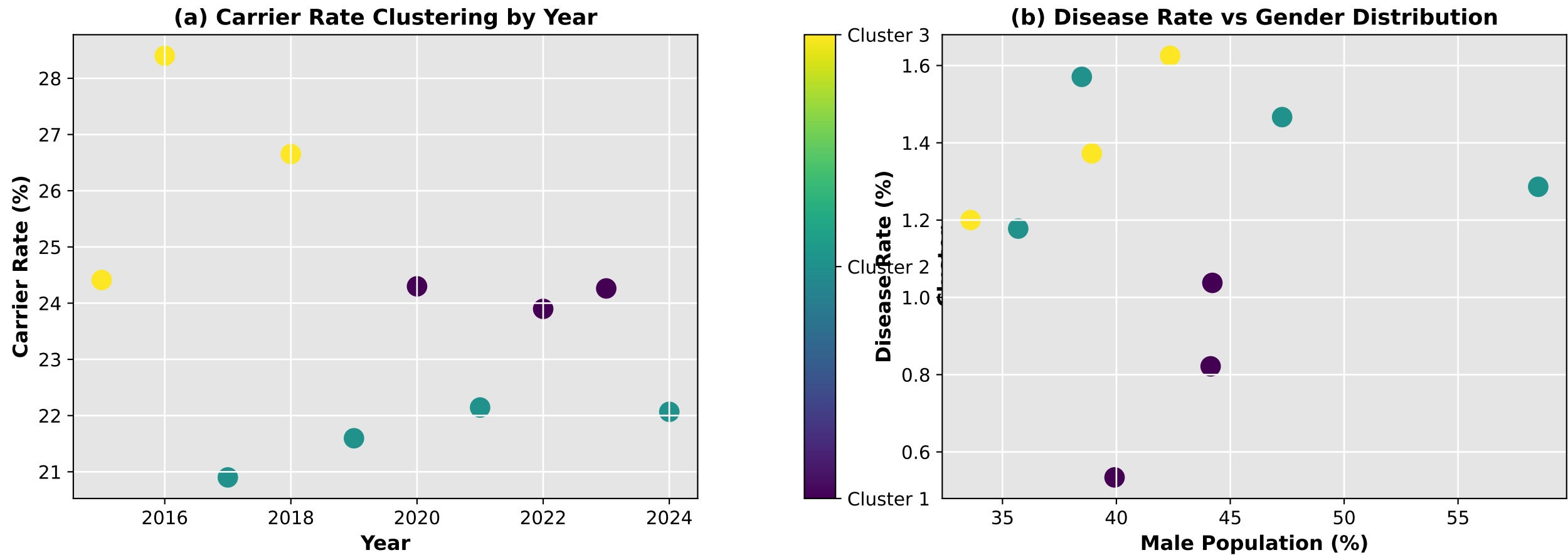


Fig. 5. Clustering Analysis of Genotype Distribution Patterns



|                             |                                             |
|-----------------------------|---------------------------------------------|
| Clustering Results Summary: |                                             |
| Cluster 1 (Stabilisation):  | Years represented: [2020, 2022, 2023]       |
|                             | Avg Carrier Rate: 24.15%                    |
|                             | Avg Disease Rate: 0.80%                     |
|                             | Avg Male Rate: 42.76%                       |
| Cluster 2 (Transition):     | Years represented: [2017, 2019, 2021, 2024] |
|                             | Avg Carrier Rate: 21.68%                    |
|                             | Avg Disease Rate: 1.38%                     |
|                             | Avg Male Rate: 44.99%                       |
| Cluster 3 (High-burden):    | Years represented: [2015, 2016, 2018]       |
|                             | Avg Carrier Rate: 26.49%                    |
|                             | Avg Disease Rate: 1.40%                     |
|                             | Avg Male Rate: 38.29%                       |